

Day 06 — Data Preprocessing & Feature Engineering

GitHub-ready theory notes

Completed: 12 / 19 topics

1. Why Data Preprocessing is Needed

Real-world data is messy: missing values, categorical values, different scales, outliers, and leakage risks. Data preprocessing converts raw data into a form suitable for ML. Typical flow: raw data → missing values → encoding → scaling → outlier handling → leakage prevention → model.

2. Train / Validation / Test Split

Training: model learns parameters. **Validation:** tune hyperparameters and choose models. **Test:** final evaluation on unseen data. Common splits include 70/15/15 or 80/20 train/test. For small datasets, cross-validation is useful. Split before fitting preprocessing transformations.

3. Data Leakage

Leakage occurs when information that should not be available during training enters the training process. Example: using future information to predict a target. Golden rule: anything that learns from data must be fitted using training data only — imputer, scaler, feature selection, PCA, etc. Pattern: fit on train → transform train/validation/test.

4. Advanced Missing-Value Handling

Numerical: mean for roughly symmetric data without major outliers; median for skewed data or outliers. Categorical: mode. Time series: forward/backward fill when logically appropriate. KNN uses similar rows to estimate missing values. Iterative/MICE predicts missing values using other features. Missing indicators can capture informative missingness. Drop rows/columns only when justified. MCAR = completely random; MAR = depends on observed information; MNAR = depends on unobserved/missing information. Fit imputation statistics on training data only.

5. Categorical Encoding

Categorical features contain categories such as City, Gender, Color, or Size. Encoding converts categories into numerical representations. **Nominal:** no meaningful order. **Ordinal:** meaningful order. Avoid arbitrary numeric labels for nominal categories because they can create a fake order.

6. One-Hot Encoding

Used mainly for nominal categories with no natural order. Example: City = Hyderabad, Chennai, Vizag becomes three binary columns: Hyderabad → [1,0,0], Chennai → [0,1,0], Vizag → [0,0,1]. This avoids artificial ordering.

7. Label Encoding vs Ordinal Encoding

Binary categories can naturally be represented as 0/1, such as Yes/No. For multi-category nominal input features, One-Hot Encoding is generally preferred. For ordered categories, Ordinal Encoding is appropriate: Beginner → 1, Intermediate → 2, Advanced → 3. In scikit-learn, LabelEncoder is primarily intended for target labels (y); for input features (X), OneHotEncoder or OrdinalEncoder is generally more appropriate.

8. Feature Scaling

Scaling brings numerical features to comparable scales. It matters especially for KNN, K-Means, SVM, PCA, and many gradient-based models. Decision trees and tree ensembles generally do not require scaling because they use threshold-based splits.

9. Standardization

Standardization transforms a feature to mean approximately 0 and standard deviation approximately 1. Formula: $z = (x - \mu) / \sigma$. In sklearn: StandardScaler. Fit only on training data, then transform train and test using the same fitted scaler.

10. Normalization (Min-Max Scaling)

Min-Max scaling usually maps values to 0–1. Formula: $x' = (x - x_{\min}) / (x_{\max} - x_{\min})$. Minimum becomes 0 and maximum becomes 1. In sklearn: MinMaxScaler. It is sensitive to outliers because it depends on min and max.

11. Robust Scaling

Useful when outliers are present. It uses median and IQR, so extreme values have less influence. Formula: $x' = (x - \text{median}) / \text{IQR}$, where $\text{IQR} = Q3 - Q1$. In sklearn: RobustScaler. It does not remove outliers; it only makes scaling less sensitive to them.

12. Outliers

An outlier is a value unusually far from most observations. Outliers can distort mean, standard deviation, scaling, and some ML algorithms. IQR method: $\text{IQR} = Q3 - Q1$; lower bound = $Q1 - 1.5 \times \text{IQR}$; upper bound = $Q3 + 1.5 \times \text{IQR}$. Values outside are potential outliers. Box plots help visualize them. Detection does not mean automatic removal. First decide whether the value is valid, unusual but meaningful, or an error. Possible actions: keep, remove, cap/winsorize, transform, or use robust methods.

Quick Revision Table

Technique	Main idea	Outlier sensitivity
Standardization	Mean + Standard Deviation	Sensitive
Min-Max	Min + Max; usually 0–1	Very sensitive
Robust Scaling	Median + IQR	More robust
One-Hot	Binary column per nominal category	N/A
Ordinal	Numbers preserve meaningful order	N/A

Day 06 status: 12/19 completed. Remaining: Feature Selection, Feature Engineering, Pipelines, ColumnTransformer, Imbalanced Data, SMOTE, and End-to-End Preprocessing Pipeline.